MILITARY SPECIFICATION

CLOCK, AIRCRAFT, ELECTRONIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quartz time base, electronic, aircraft clocks which shall indicate hours, minutes and seconds for clock time and elapsed time modes in a six digit, twenty-four hour numeric display in addition to a sweep second, analog indicator. The clock shall include a keep alive timing function. Clocks shall contain provisions for both blue filtered white lighting and red lighting in a dual integral lighting system, or shall contain provisions for Night Vision Goggle (NVG) compatible green lighting.

1.2 Part numbers. The following part numbering system shall be used to specify the clocks:

M or D 87979
- 1 Blue or Red Light, Liquid Crystal Numerals
- 2 Blue or Red Light, Emitting Numerals
- 3 Green Night vision, Liquid Crystal Numerals
- 4 Green Night vision, Emitting Numerals

\- Specification Number

M - Inch-Pound (US Units)
D - Metric Units

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-585	Box, Wood Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6645

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PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Shipping Fiberboard
QQ-S-781	Strapping, Steel, and Seals

MILITARY

MIL-P-116	Preservative, Methods of
MIL-T-7788	Panels, Information, Integrally Illuminated
MIL-P-11268	Parts Materials, and Processes Used in Electronic Equipment
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges.
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification for
MIL-L-85762	Lighting, Aircraft, Interior, AN/AVS-6 Aviator's Night Vision Imaging System (ANVIS) Compatible

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized Unit Loads
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measuring of
MIL-STD-704	Aircraft Electric Power, Characteristics
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS27466	Connector, Receptacle, Electrical Wall Mounting Flange, Crimp Type, Bayonet Coupling Series 1.

2.1.2 Other Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOT-E-7052 49 CFR (Code of Federal Requirements)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951	Packaging, Commercial

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Copies of specifications, standards, handbooks, drawings, publications,and other Government Documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electronic clocks under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Nonmagnetic materials. Nonmagnetic parts shall be used for all parts of the electronic clock, except when magnetic parts and ferrous materials are essential.

3.1.2 Toxic and corrosive fumes. Materials as installed in the electronic clock and under the service conditions specified herein, shall have no adverse effect on the health of personnel when used for its intended use. This shall include any fungicidal agents that are used.

3.1.3 Protective treatment. When materials are used, in the construction of the electronic clock, that are subject to deterioration and corrosion when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against any such deterioration in a manner that will in no way prevent compliance with the requirements specified herein. Protective coatings that will crack, chip, or scale during extremes of climatic or environmental conditions shall not be used.

3.1.4 Fungusproof materials. Materials that are nutrients for fungus shall not be used unless no other material can be used for that application. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.1.5 Metals. Metals shall be of the corrosion resistant type, unless protected to resist corrosion during normal service life.

3.1.5.1 Dissimilar metals. Dissimilar metals are defined in requirement 16 of MIL-STD-454. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided.

3.1.6 Protection of ferrous materials. Ferrous materials contained within hermetically sealed enclosures shall be considered suitably protected from corrosion.

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3.1.7 Standard parts. The number of different types of parts shall be held to a minimum. Control of the selection and testing of all parts shall be in accordance with MIL-P-11268.

3.2 Design and construction.

3.2.1 Case. The size and configuration of the case shall be as shown in Figure I. The case shall consist of a body and a front panel. The case shall withstand normal shipping, handling and installation procedures. The case shall not be crushed by normal torque applications of 12 inch - pound by mounting screws.

3.2.1.1 Case finish. The case shall be uniform in texture, and shall have a smooth surface with a durable lusterless black finish. The finish shall not present evidence of breakdown during or after testing. Trademarks, company names, or other markings not related to the panel function shall not be displayed on the panel face.

3.2.1.2 Touch temperature. The maximum stabilized surface temperature of the electronic clock front panel and controls shall not exceed 49°C with a cockpit temperature of 26°C.

3.2.2 Panel lighting. If needed, th panel lighting shall operate on 5 VAC and 28 VDC or 115 VAC and shall conform to MIL-P-7788. The brightness shall be controlled by the aircraft dimmer control.

3.2.3 Thermal design. The clock shall be designed to operate throughout the temperature range of 3.3.1 without requiring forced air or liquid cooling.

3.2.4 Electrical connector. The electrical connector shall be located as shown in Figure 1. The connector used shall be MS27466T9B35PN in accordance with MIL-STD-454. The wiring shall conform to Figure 2.

3.2.4.1 Safety ground. A wire of minimum length connected internally to the equipment chassis shall be provided at pin number five on the power connector. No circuit shall be allowed to utilize this wire as its return.

3.2.5 Clock system. The solid state electronic clock shall contain: (1) A module which generates the required master frequency and controls the accuracy; (2) Solid state counters used to keep track of time in the clock and elapsed time modes; (3) A six digit twenty-four hour numeric display with fixed colons between the hour; minute and second places; (4) A sweep second indication. The clocks shall meet the following criteria:

a. Clocks with part numbers -1 and -2 shall contain provisions for dual intergral lighting. Blue filtered white lighting shall be in accordance with MIL-L-27160, and red lighting in accordance with MIL-L-25467, as applicable. Clocks with part numbers -3 and -4 shall have Night Vision Goggle (NVG) compatible green lighting in accordance with MIL-L-85762. The display shall be such that it is clearly readable under ambient light conditions. If the display is designed to be seen in daylight without internal illumination the night lighting shall operate on 5 VAC and 28 VDC or 115 VAC and shall be in accordance with MIL-L-27160 or MIL-L-25476. If the display uses light emitting numerals for daylight viewing it shall operate from 28 volts dc.

b. Brightness control - A front mounted brightness control shall be provided for clocks with use of 28 volts dc for energizing light emitting

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numerals. The control shall provide approximately linear, uniform, diming characteristics from full bright to off. The lighting must not drop out before .03 footlamberts. Those clock displays which use separate night lighting shall operate from the aircraft instrument brightness control.

a. Contrast - When at maximum intensity, the lighted display segments shall provide a contrast ratio of not less than 2 in a diffuse ambient environment of 10,000 foot-candles measured at the face of the display. Unactivated display segments shall demonstrate a contrast ratio no greater than 0.1 in a diffuse ambient of 10,000 foot-candles.

$$\text{Contrast ratio} - \frac{B_2 - B_1}{B1}$$

Where B_1 = background brightness

B_2 = display segment brightness

Display segment brightness shall be the average of at least three(3) spot measurements equally spaced from each other including one at each end of the segment and at the center of the segment.

The perceived luminance (Lp) shall be at least equal to the quantity below across the entire ambient light spectrum of 10^{-4} to 10^{6} footlamberts.

$$Lp \geq 3.87B_1 \ (.926) + .05$$

Where $Lp = B_2 - B_1$

d. Display uniformity - The numerals and segments shall at all times appear to be uniform in size and color. The apparent size of the segments or numerals shall not appear to change over the entire intensity range. The average brightness of each segment shall be determined by taking a minimum of three spot measurements, one at the center of the segment and one at each end of the segment. The maximum and minimum spot brightness within the segment shall be limited such that the brightest spot shall be not more than twice the brightness of the least bright spot. This shall be tested with the lighting voltage of 4.5$\pm$ .5 VAC, 28.0 $\pm$ 2.8 VDC. The brightness variation between each segment of each numeral shall not be greater than 1.3 times nor less than .7 times the average brightness of all segments in the numerals when measured at 4.5 V $\pm$ .5 or 115 $\pm$ 11.5 VAC. The brightness variation between each numeral of the display shall not be greater than 1.2 times nor less than .8 times the average brightness of all numerals in the display when measured at 4.5 $\pm$ .5 VAC.

Where: Average numeral brightness is the average of all 7 segment spot readings (21).

Average segment brightness is the average of all segment spot readings (3).

e. The individually lighted segments shall not reduce their apparent length when the brightness is reduced to night level conditions.

f. The brightness control, if needed, shall be continuous with an extended range for night time brightness. At low levels of brightness, individual segments shall not drop out of view.

g. All transparent reflection surfaces of the segmented readout display

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shall be provided with a reflection reducing coating that meets the requirements of MIL-C-14806.

h. The digits shall be sharp and clearly readable in all ambient conditions. There shall be a minimum viewing cone of 60°. The digits shall not subtend less than 7 mrad (24 min) of visual angle. The minimum viewing distance shall be 28 inches. The analog sweep second indicator shall conform to MIL-STD-1472, paragraph 5.2.3.1.7 and paragraph 5.2.3.2. With a 10,000 foot-candle light source positioned anywhere from normal to 80 degree from the display face, the display shall be clearly readable. Note: "Clearly Readable" shall mean that anywhere in the viewing cone, at a distance of 36 inches, the correct time can be read by the operator within 3 seconds.

i. An italic or sloping stroke mosaic may be used, but the slope shall not exceed 11° from vertical. If sloping characters are used, a slant of 5° to 7° is preferred.

3.2.6 Clock controls. All clock controls shall be clearly labeled and visible in all operating conditions. Labels shall be in accordance with MIL-STD-1472, paragraph 5.5.

3.2.6.1 Mode selection. The display shall have two modes, one for clock time and one for elapsed time. The electronic clock shall have a select push button to choose between the clock and elapsed time modes. Pressing this button momentarily shall change the clock mode. The display shall indicate which mode the clock is in at any time by showing a "C" when in clock time and "ET" for elapsed time mode, below the display. When power is first applied the clock shall come up in the clock time mode.

3.2.6.2 Self test. If the display is designed for illumination during daylight, the brightness control shall have a push-to-test function which shall cause all segments to illuminate. This test shall not interrupt the time keeping function of the clock.

3.2.6.3 Control buttons. There shall be clearly labeled control push buttons to function as described below. The "select" button shall be on the left side of the clock and the "control" button shall be on the right side. The buttons shall be designed to allow easy use by a crewmember wearing flight gloves. A positive indication of control activation shall be provided, such as a snap feel. The push button surface should be normally concave (intended) to fit the finger. When this is impractical, the surface shall provide a high degree of frictional resistance to prevent slipping. The button dimensions, resistance, displacement, separation shall conform to Figure II of MIL-STD-1472. The buttons shall be located on the lower portion of the clock face such that when they are being operated they shall not block the view of the display.

3.2.6.4 Time set. With the clock mode, set the time by first depressing the "select" and "control" buttons "simultaneously." For example, if the first button is depressed and held an instant before the second button is depressed, then the clock will still go into the time-setting mode. This activates the display for setting with the first digit flashing, awaiting setting. The update setting of this digit is accomplished by momentary depression of the "control" button. Each depression of the control button shall update a digit one unit for each depression. Depressing the "select" button sets the next digit flashing, awaiting setting. All other digits are set in the same manner.

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3.2.6.5 Elasped time controls. With the clock in the elapsed time mode, momentary depression of the "control" button shall start the elapsed timer counting upward. A second, third and fourth depression of the "control" button shall stop, zero, and start the timer provided the clock is in the elasped time mode. The analog second indicator on the clock shall be synchronized with the elapsed timer. Switching back and fourth between clock and elapsed time modes shall not affect the performance of either mode.

3.2.6.6 Instructions. Instructions for installation and operation shall be provided with each unit. (See 6.2.2).

3.2.7 Electrical power.

3.2.7.1 Clock power. The electronic clock shall be designed to operate from 28 volts dc input power in accordance with MIL-STD-704 normal and abnormal conditions. Total power consumption shall not exceed 5 watts, including lighting power, and any needed display heating. The clock shall be capable of operation, excluding display, during aircraft power interrupts and transients in accordance with MIL-STD-704.

3.2.7.2 Panel power. The panel lighting shall be designed to operate from 0.5 VAC, 0.28 VDC for incandescent lighting and 0.115 VAC for electroluminescent lighting.

3.2.7.3 Keep alive battery. A keep alive battery system shall be provided for the electronic clocks. The purpose of the keep alive battery system shall be to keep the timing circuits of the clock continuously operating for a minimum of one year when aircraft (28 vdc) power is not applied. The clocks shall be shipped without batteries. The display shall not be required to operate without the 28 vdc aircraft power applied.

3.2.7.3.1 Battery design. The battery system shall be designed so that it can easily be disconnected, removed, and replaced without special equipment or removing other components. The polarity of the battery terminals shall be prominently marked. The connections shall be designed to prevent reverse polarity engagement. The battery compartment shall be designed to prevent pressure build up from heat, gasses, liquids, or chemicals released during battery operation, charging, deterioration or rupture and shall also prevent such materials from entering the electronic compartment or cockpit. The battery circuitry shall be designed to insure that the battery cannot be charged by an outside power source. The battery circuitry shall limit the output current rate in case of an external short applied to the system. The clock circuitry shall relieve the battery load upon application of external aircraft power. A discharged or missing battery shall not prevent normal operation of the clock using aircraft power. The battery system shall not be damaged when stored in the temperature stated in 3.3.1. The keep alive system shall function from batteries providing 3 volts and 500mAH. The battery shall be a ½ AA size.

3.2.7.3.2 Installation markings. Connections, polarity, minimum acceptable voltage for operation, nominal voltage and types of batteries required shall be marked in a prominent place on or adjacent to the battery compartment.

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3.2.7.3.3 Warning label. The clock case shall be labeled externally as follows:

WARNING
Remove batteries before
shipment or inactive storage
of 30 days or more

3.2.7.3.4 Lithium batteries. If lithium batteries are used care shall be taken to meet the transportation and disposal requirements in accordance with DOT-E-7052 and 49 CFR (Code of Federal Regulations).

3.2.8 Reliability. The electronic clock shall have a lower test limit Mean-time-between-failure (MTBF) of 3000 hours, fully operating including display, at 90 percent confidence level as defined in MIL-STD-781.

3.2.9 Useful life. The clock shall have a useful life of not less than 15 years under any combination of operating and storage life, when the operational service life has not been exceeded.

3.2.10 Operational service life. The clock shall have an operational service life after delivery of not less than 10,000 hours, fully operating including display, under any natural combination of environmental conditions specified herein. Operational service life is defined as the total operating time between the start of operation and wear out. Wear out is defined as the point where overhaul or repair cost exceeds one-half of the replacement cost of the equipment.

3.2.11 Maintainability program. The following shall be considered in the design of the electronic clock:

a. Minimization of complexity of maintenance tasks (for example: Battery replacement, inspection etc.) by maximum use of simple design which induces optimum interchangeability and use of standardized equipment or commercial items.

b. Optimum accessibility in all systems, equipments and components requiring maintenance, inspections, removal, or replacement.

3.2.11.1 Service and removability. The electronic clock shall be designed to require no corrective or preventive maintenance while installed in the aircraft. The electronic clock shall be removable by removing two mounting screws, disconnecting the electrical connector and lifting the electronic clock from the aircraft instrument panel.

3.2.12 Safety. The electronic clock shall present no danger, injury, or hazard to operating and service personnel.

3.3 Performance. The electronic clock shall operate as specified herein under the following conditions:

3.3.1 Temperature. Operating temperature ranging from -40°C to +71°C. Clock storage temperatures of -73°C to +95°C.

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3.3.2 Temperature-altitude. Temperatures ranging from -40°C to +71°C at altitudes from sea level to 100,000 feet.

3.3.3 Humidity. Relative humidity up to 95 percent including condensation as defined in MIL-STD-810, Method 507, Procedure I.

3.3.4 Salt fog. Exposure to simulated salt sea atmosphere as defined in MIL-STD-810, Method 509, Procedure I.

3.3.5 Vibration. Random vibration as defined by Figure III and gunfire vibration as defined by Figure IV in accordance with MIL-STD-810, Method 514 and 519.

3.3.6 Fungus. Fungus growth as encountered in tropical climates as defined in MIL-STD-810, Method 508.

3.3.7 Dust. Sand and dust particles as encountered in desert areas as defined in MIL-STD-810, Method 510.

3.3.8 Shock. Shock forces as described in MIL-STD-810, Method 516, Procedure I and Procedure III.

3.3.9 Acceleration. Acceleration as described below in accordance with MIL-STD-810, Method 513, Procedures I and II. (See 4.2.23.2).

Procedure I	Procedure II
Structural g levels	Operational g levels
Fore 9.0	Fore 2.0
Aft 9.0	Aft 6.0
Up 13.5	Up 12.0
Down 4.5	Down 3.0
Lateral 6.0	Lateral 4.0

3.3.10 Temperature shock. Exposure to 71°C, then to -40°C, and return to 71°C under the conditions specified in MIL-STD-810, Method 503.

3.3.11 Electromagnetic interference. The clock shall neither generate nor be susceptible to electromagnetic interference. Electromagnetic interference shall be in accordance with MIL-STD-461 as specified for Class A1 equipment. The specific methods required are: CE03, CS01, CS02, CS06, RS02, and RS03.

3.3.12 Dielectric strength. There shall be no insulation breakdown when 250 volts dc is applied between isolated pins and between pins and the case for a period of 10 seconds.

3.3.13 Error. Error shall not exceed the tolerance specified in Table I.

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3.3.14 Sunshine. Sunshine as encountered in normal operating conditions as defined in MIL-STD-810, Method 505.

TABLE I. Error tolerances.

Accuracy per 24 hrs. (room temperature)	± 1 second
Accuracy during low temp operation for 24 hrs.	± 4 seconds
Accuracy during high temp operation for 24 hrs.	± 4 seconds
Accuracy of elapsed time mode per 24 hrs. (room temperature)	± 1 second

3.4 Weight. The weight of the clock shall not exceed 16 ounces including battery.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Workmanship. The electronic clock shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

3.6.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. The clocks must be interchangeable and compatible with the existing instrument panel cut out.

3.6.2 Screw assemblies. Assembly screws and bolts shall be "tight." The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt threads, or assembly.

3.6.3 Packaging. The clocks shall be packaged in accordance with section 5 of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.1 Tests. All tests shall be run in the listed order.

4.1.2 Individual acceptance tests. Each electronic clock shall be subjected to and shall pass the following:

a. Examination of product.

b. Accuracy

c. Lighting

d. Select switch operation.

e. Dim control operation.

f. Elapsed time operation.

g. Time set operation.

h. Burn in test.

4.1.2.1 Qualification testing.

4.1.2.2 Test samples. The qualification test samples shall consist of at least seven clocks representative of the production equipment. Three of the samples shall be subjected to all the tests in sampling plan A and B, while seven samples (may include the three subjected to the sampling plan tests) shall be subjected to the sampling plan C tests. Upon completion of the qualification tests the individual acceptance tests, excluding the burn in test, shall be run again to assure the clock functions properly.

4.1.2.3 Sampling plan A. A minimum of 3 clocks shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests:

a. Individual tests (see 4.1.1).

b. Low temperature.

c. High temperature.

d. Extreme voltage variation.

e. Dielectric strength.

f. Vibration operation.

4.1.2.4 Sampling plan B. Unless otherwise specified, 3 clocks shall be selected at random from the first 10 items produced on each contract or order and subjected to the following tests in the order shown. If the clock case is hermetically sealed the dust, humidity, fungus and salt fog tests may be performed on the empty case.

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a. Sampling plan A tests (see 4.1.2.2.).

b. Temperature shock.

c. Temperature altitude.

d. Humidity

e. Salt fog.

f. Fungus

g. Dust

h. Electromagnetic interference.

i. Vibration

j. Gunfire vibration.

k. Acceleration

l. Shock

m. Sunshine

4.1.2.5 Sampling plan C. Seven clocks shall be selected at random from the first 14 produced and subjected to the following tests.

a. Individual acceptance (see 4.1.1).

b. Reliability

4.1.3 Rejection and retest. When an item selected from a production run fails to meet the specification due to a relevant failure, meaning failure of a mission essential capability, no items still on hand or later produced shall be accepted until the extent and cause of failure has been determined and appropriately corrected. The contractor shall provide descriptive documentation to the Government representative about the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.1.4 Standard conditions. The following conditions shall be used to establish normal functional performance characteristics:

a. Temperature: $25^{o} \pm 10^{o}C$.

b. Surrounding air pressure: Room ambient conditions.

c. Humidity: Room ambient up to 90 percent relative humidity.

d. Lighting level: Maximum brightness.

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4.1.5 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.2).

4.1.6 Erratic operation. After completion of and during the tests specified, the clock shall be operated and inspected. No signs of abnormal operation or physical damage shall be present.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality of conformance inspection (see 4.4).

4.2.1 Examination of product. The electronic clock shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking. The electronic clock shall be placed in the face-down position, shaken by hand, and inspected in this position to determine that no loose hardware or foreign materials are present.

4.2.2 Accuracy. The electronic clock shall be electrically connected and powered. It shall be operated for a minimum of 240 hours and shall not exceed the tolerances for error shown in Table I.

4.2.3 Lighting and display.

4.2.3.1 Panel lighting. The panel shall be tested per MIL-P-7788 as applicable, in accordance with the requirements cited herein.

4.2.3.2 Display. A test shall be performed to verify compliance with requirements of 3.2.5 (a, b, c, d, f, g, h, and i).

4.2.4 Mode select switch operations. With the electronic clock properly connected and powered the "select" switch shall be set in the clock mode and then in the elapsed time mode. In each position the electronic clock shall operate in the correct corresponding mode.

4.2.5 Dim control operations. With the electronic clock properly connected and powered, the dim control knob, if provided, shall be operated. The extremes of the dim control should cause the display to go from full bright to completely off. During this test all display requirements listed in 3.2.5 shall be met. The dim control shall then be depressed and cause all segments to illuminate.

4.2.6 Elapsed time operation. With the electronic clock properly connected and powered, it shall be operated in the elapsed time mode for a minimum of 24 hours. The error tolerances of Table I shall not be exceeded.

4.2.7 Time set operation. With the electronic clock properly connected and powered the time setting controls shall be operated to confirm compliance with 3.2.6.4.

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4.2.8 Burn in test. The clock shall be operated for 24 hours at a temperature of 50° C. During and after the test the clock shall show no signs of erratic operation.

4.2.9 Low temperature. The electronic clock shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502. The electronic clock shall show no evidence of erratic operation.

4.2.10 High temperature. The electronic clock shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501. There shall be no evidence of erratic clock operation. Thermal evaluation and testing shall verify compliance with the requirements of 3.2.1.2 and 3.2.3.

4.2.11 Extreme voltage variation. The electronic clock shall be properly connected and, with input power applied and set at extreme voltages as specified in MIL-STD-704, Table II, undervoltages, overvoltages and transients shall be applied in accordance with MIL-STD-704. The clock shall show no signs of erratic operations.

4.2.12 Dielectric strength. The electronic clock shall be subjected to a dielectric strength test and 250 Vdc shall be applied for a period of 10 seconds during the test between isolated pins and the case. There shall be no evidence of insulation breakdown.

4.2.13 Vibration operation. The vibration operation (airworthiness) test shall be conducted in accordance with MIL-STD-810, Method 514, for 1/2 hour per axis using the performance test spectrum of Figure V of this specification. The clock shall show no signs of erratic operation.

4.2.14 Temperature shock. The electronic clock shall be subjected to a temperature shock test in accordance with MIL-STD-810, Method 503, Procedure I, except that the test shall be continued for eight cycles. After the electronic clock has returned to room temperature, it shall be examined and there shall be no evidence of any cracked connectors, circuit or module breaks. The electronic clock shall then be operated and show no sign of erratic operation.

4.2.15 Temperature - altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, Method 504, Procedure I, Category 7. The electronic clock shall be connected with no power applied. Pressure in the chamber shall be reduced to the equivalent of 100,000 feet ±500 feet and power shall be applied with the temperature limits as stated in 3.3.1. The electronic clock shall operate immediately and its operation shall be observed for 1 hour. There shall be no evidence of erratic performance . Pressure in the test chamber shall be increased to atmospheric and after the electronic clock has returned to room temperature, it shall show no signs of abnormal operation after 6 hours of operation.

4.2.16 Humidity. The electronic clock shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I, after which it shall show no signs of erratic operation.

4.2.17 Salt fog. The electronic clock shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509, Procedure I. At the end of the 48 hour period, the electronic clock shall be operated for six hours showing no signs of erratic operation.

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4.2.18 Fungus. The fungus test shall be conducted in accordance with MIL-STD-810, Method 508, Procedure I. Upon completion of this test the electronic clock shall show no signs of erratic operation. There shall be no deterioration nor shall any part of the electronic clock support fungus growth.

4.2.19 Dust. The dust test shall be conducted in accordance with MIL-STD-810, Method 510, Procedure I. Upon completion of this test the electronic clock shall be examined to determine that no physical deterioration has occurred. The clock shall show no signs of erratic operation.

4.2.20 Electromagnetic interference. The clock shall be tested in accordance with the applicable requirements of MIL-STD-462, to verify compliance with 3.3.11. The electronic clock shall not experience erratic operation.

4.2.21 Vibration. The electronic clock shall be tested in accordance with MIL-STD-810, Method 514 using the spectra of Figure 3 of this specification. The clock shall not exhibit degraded performance or erratic operation during vibration per the performance curve not after vibration per the endurance curve. The endurance test shall be 1 hour in each axis.

4.2.22 Gunfire vibration. The electronic clock shall be tested in accordance with MIL-STD-810, Method 519 using the test described herein. The electronic clock shall not experience erratic operation during or after the test. The test shall consist of 15 minutes of random testing per axis according to the spectrum of Figure 4 of this specification. In addition, the clock will be subjected to a 7 1/2 minute sine dwell per axis at the following acceleration levels and frequencies.

Frequency Hz	Acceleration g
33	2.5
67	5.0
100	5.0
133	5.0
167	5.0
200	5.0
233	5.0
267	5.0

4.2.23 Acceleration. The acceleration tests shall be performed in accordance with MIL-STD-810, Method 513, Procedures I and II, except where noted herein. There shall be no signs of erratic clock operation.

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4.2.23.1 Structural test. The electronic clock shall be subjected to the acceleration specified in 3.3.9 for the appropriate axes in accordance with MIL-STD-810, Method 513, Procedure 1, for a period of one minute. The fore direction shall be defined with the electronic clock mounted in its normal operating position and the rear of the case toward the center of rotation.

4.2.23.2 Operational test. The electronic clock shall be mounted in each of the three positions specified in MIL-STD-810, Method 513 and subjected to the operational accelerations specified in 3.3.9 for the appropriate axes.

4.2.24 Shock. The electronic clock shall, while operating, be subjected to shock test in accordance with MIL-STD-810, Method 516, Procedure I basic design and Procedure III crash safety. The saw tooth wave shall be used. The electronic clock shall not be damaged nor shall it subsequently fail or show erratic operation as a result of this test.

4.2.25 Sunshine. The sunshine test shall be performed in accordance with MIL-STD-810, Method 505, Procedure II for 3 cycles of 24 hours.

4.2.26 Reliability. The electronic clock shall be subjected to the reliability test in accordance with MIL-STD-781, test plan IIIC and Table I, which shall demonstrate a lower test limit MTBF of 3000 hours. The test shall incorporate Categories 4 and 5 as described in MIL-STD-781, Appendix B. Each clock shall run a minimum of one half the average test time based on the total time of at least seven units. The test chamber shall be temperature cycled for 2 hours, stabilized at the low operating temperature ($-40^{o}C$), followed by 2 hours, stabilized at the high operating temperature ($+71^{o}C$), for the test duration in accordance with MIL-STD-781. During the test the clocks shall be subjected to sine vibration at 2.2G $\pm$ 10% peak acceleration level at any nonresonant frequency between 20 and 60 cps. The duration of the vibration shall be at least 10 minutes during each hour of clock operating time. Each electronic clock shall be cycled from elapsed time mode to clock mode every 24 hours. Once every 24 hours, the dim control knob, if supplied, shall be fully rotated (clockwise and counter clockwise) ten times and depressed ten times. Every 200 hours the individual tests, excluding burn in test, lighting tests and with only a 24 hour accuracy test, shall be run to determine whether any relevant failures have occurred. Acceptance or rejection shall be determined in accordance with MIL-STD-781, Test Plan IIIC.

4.2.27 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of Section 5 of this specification.

4.3 Qualification inspection.

4.3.1 Qualification samples. The samples shall consist of 10 clocks representative of the production clock. Samples shall be identified with the manufacturer's part number and shall be accompanied by such other information as required. They shall be tested under the conditions specified herein and at the location designated by the activity responsible for qualification (see 6.3).

4.4 Quality conformance inspection. Quality conformance inspections shall consist of the examination, individual tests, and Sampling Plans A and B tests.

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4.4.1 Individual tests. Each clock shall be subjected to the following tests.

Tests	Applicable paragraph
Examination of product	4.2.1
Operation of elapsed time function	4.2.6
Panel lighting	4.2.3.1
Reliability acceptance	4.2.26

4.4.2 Sampling plans A and B tests.

4.4.2.1 Sampling plan A. Unless otherwise specified by the procuring activity, 4 clocks shall be selected at random from the first 20 clocks of the initial production run and, having passed the individual tests (4.5.1), shall be subjected to the following tests:

Tests	Sample No. 1 (2 clocks)	Sample No. 2 (2 clocks)	Applicable paragraph
High temperature	-	X	4.2.10
Low temperature	-	X	4.2.9
Humidity	X	-	4.2.16
Vibration	X	-	4.2.21
Operation in a magnetic field	X	-	4.2.20

4.4.2.2 Sampling plan B. Four clocks shall be selected at random from every 50, or fraction thereof, manufactured on the production run and shall be subjected to the following tests:

Test	Applicable paragraph
Burn in test	4.2.8
Gunfire vibration	4.2.22
Operational test	4.2.23
Vibration error	4.2.13
Shock	4.2.24
Lighting	4.2.3.1

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4.4.3 Rejection and retest. When one clock selected from the production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.4.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the clocks on hand or later produced shall not be made until it is determined that all clocks meet all the requirements of this specification.

5. PACKAGING

5.1 Preservation - packaging. The clock shall be packaged in accordance with MIL-STD-794, Level A, B, or C as specified in the contract. For Level A packaging, the method of preservation shall be in accordance with MIL-P-116.

5.1.1 Level C. The Level C preservation for the electronic clock shall conform to the MIL-STD-794 requirements for this level.

5.1.2 Commercial. The commercial preservation of the electronic clock shall be in accordance with the requirements of ASTM-D-3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial, as specified (see 6.2).

5.2.1 Level A. The packaged electronic clock shall be packed in containers conforming to PPP-B-601, overseas type; PPP-B-621, Class 2 Style 4 or PPP-B-585, Class 3, Style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I or IV, Finish A. When the gross weight exceeds 200 pounds or the container length and width is 48" X 24" or more and the weight exceeds 100 pounds, 3 X 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.2.2 Level B. The packaged electronic clock shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with waterproof packaging tape or box closure and sealing tape, two inches minimum width.

5.2.3 Level C. The Level C packing for electronic clock shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. The preserved electronic clock shall be packed in accordance with the requirements of ASTM-D-3951.

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5.2.5 Palletized loads. When specified, palletized loads commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer (see 6.2).

5.3 Marking levels A, B, C, and commercial. In addition to any special or other identification marking required by the contract, each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable, with the following precautionary marking appearing conspicuously on two opposite sides of each interior package and each shipping container:

FRAGILE - DELICATE INSTRUMENTS
HANDLE WITH CARE

5.4 General.

5.4.1 Exterior containers. Exterior containers shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable. (See 5.2.1, 5.2.2, 5.2.3, and 5.2.4).

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.2.27.

6. NOTES

6.1 Intended use. The clocks are intended for use as primary timepiece with an elapse time capability in aircraft requiring an integrally lighted clock.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Voltage requirements. (see 3.3.2, 3.2.5, and 3.2.7.2).

c. Selection of applicable levels of preservation, packaging, packing, marking and commercial|industrial (see 5.1, 5.2, and 5.3).

d. Part number, numeral color, and background color.

e. When palletized loads are required (see 5.4).

f. Selection of applicable levels of vibration spectrum (see 3.3.5, 4.2.13, and 4.2.21).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Engineering Support Section, SA-ALC/MMEDO, Kelly AFB, TX 78241-5000 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (keyword) listing.

Clock

Clock, aircraft

Clock, aircraft, electronic

Clock, electronic

Clock, military

Custodian:
Air Force - 99
Army - AR
Navy - AS

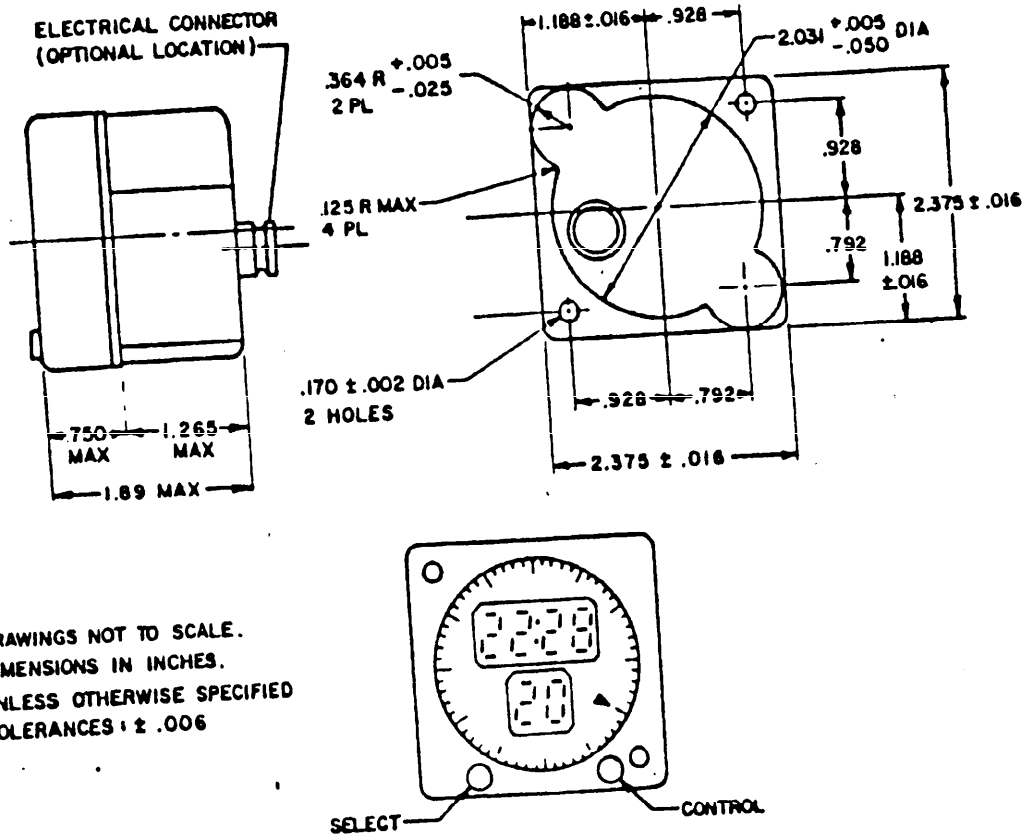
Preparing activity:
Air Force - 82

(Project 6645-0374)

Reviewing activity:

DLA-GS

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NOTES:

1. DRAWINGS NOT TO SCALE.
2. DIMENSIONS IN INCHES.
3. UNLESS OTHERWISE SPECIFIED TOLERANCES: ± .006

FIGURE 1. Clock

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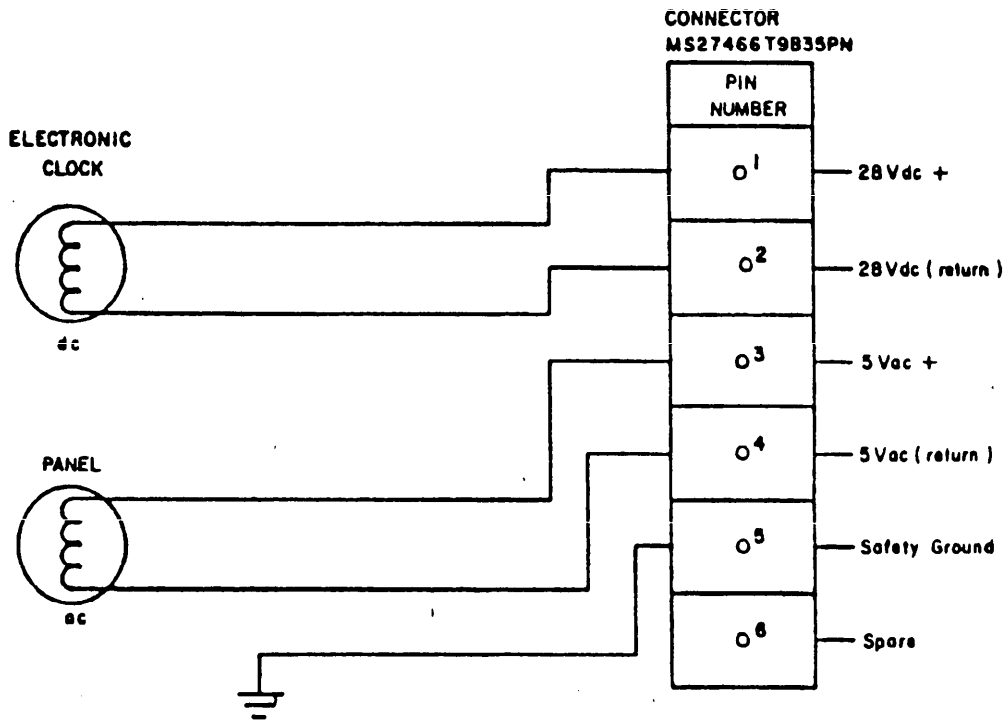


FIGURE 2. Wiring

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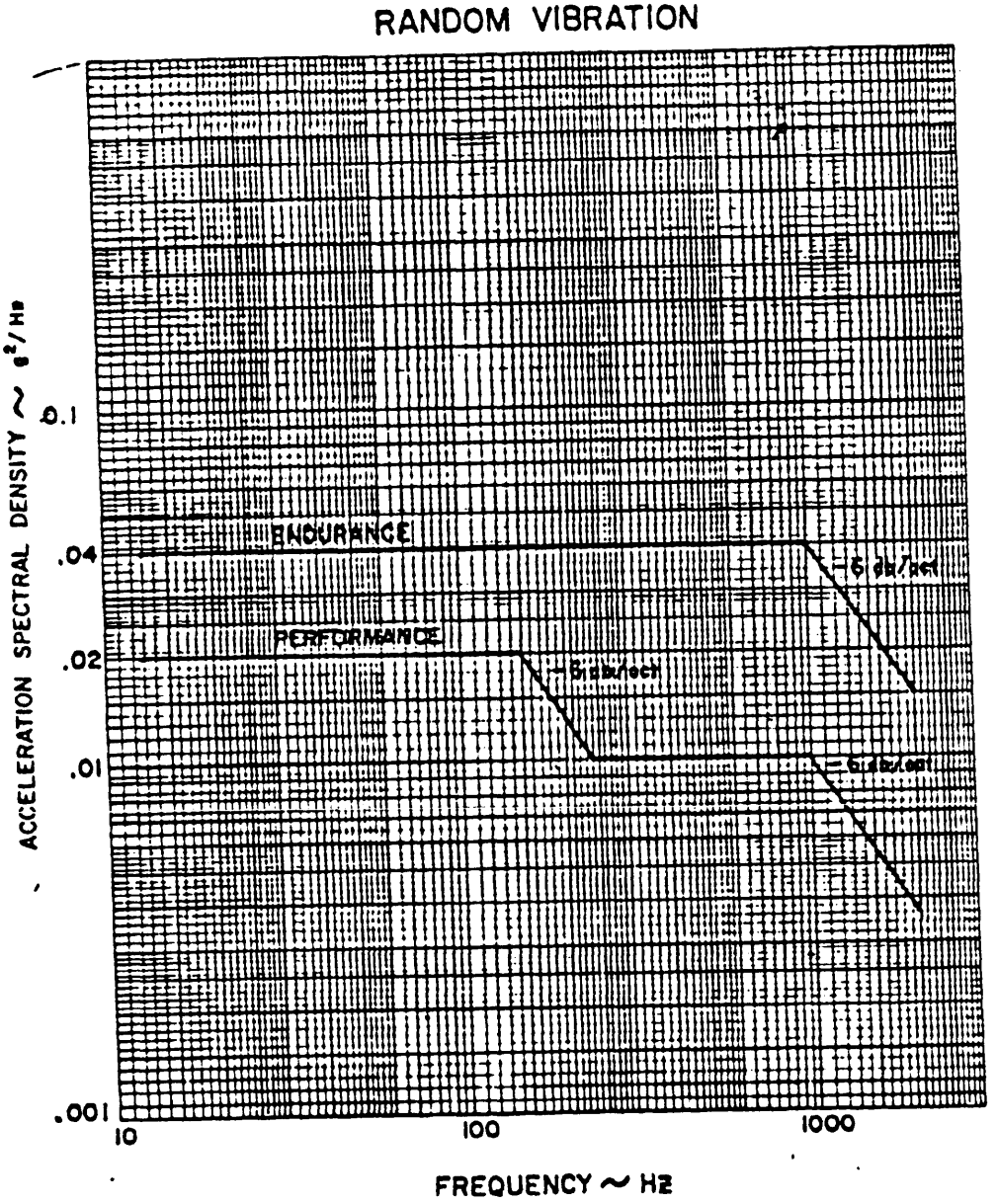


FIGURE 3.

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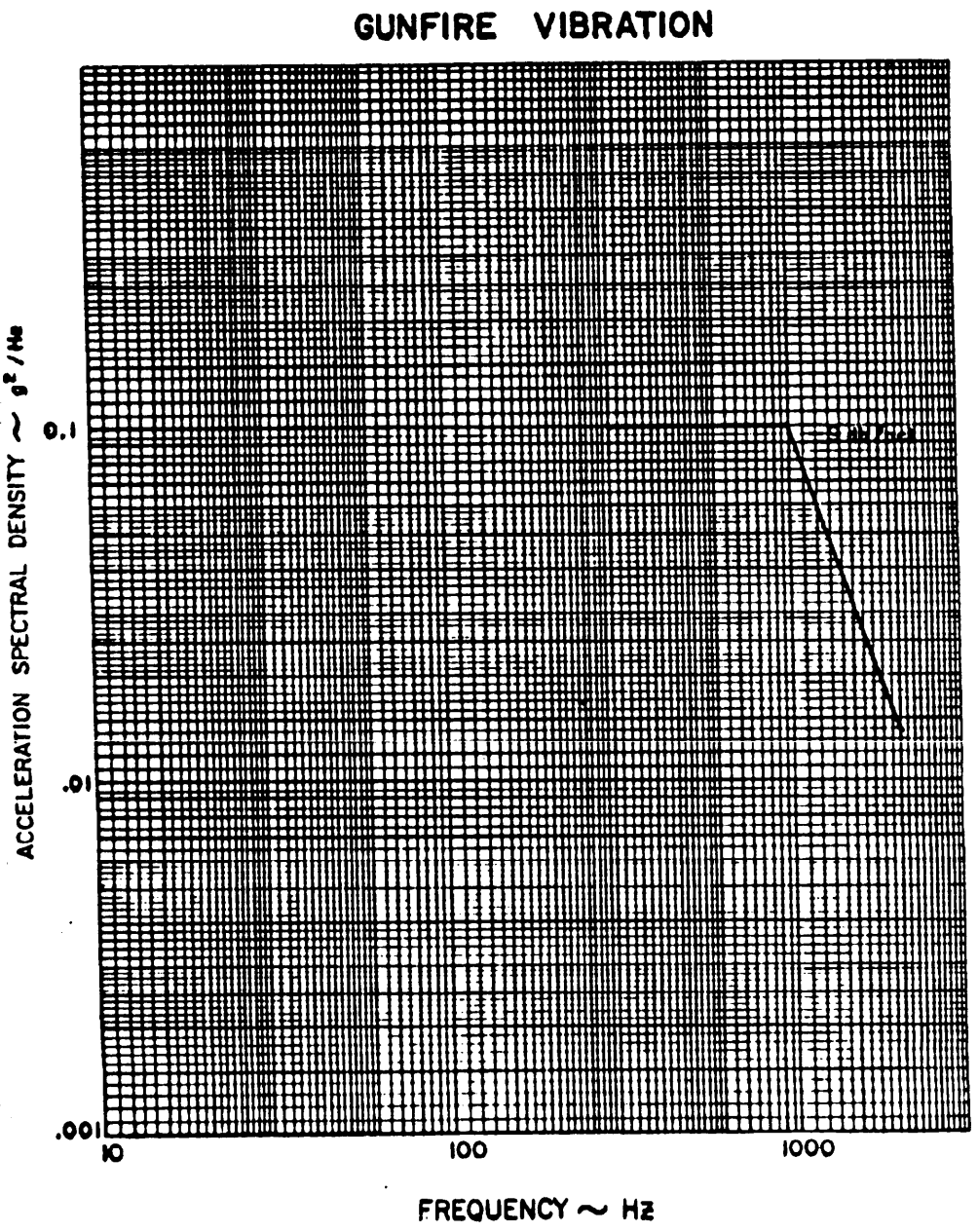


FIGURE 4.

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